Template

Studentnames and studentnumbers here

2026-06-17

Set-up your environment

Title Page

Ilias Abgij, Denis Bykov, Finn van Echtelt, Merlijn Haster, Milan Kroes, Luuk Neefjes, Raluca Oprina

Tutorial group 04

Fatima Amankour

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The relation between wealth and happiness in countries is a social problem because many societies think that wealth is the key to a happy life. Research shows that happiness is associated with wealth, particularly when basic needs are not met. However, studies also suggest that beyond a certain point more wealth does not equal higher happiness. So the question is, can economic wealth on it's own improve quality of life?

Despite all this research, gaps remain. Researchers continue to debate whether wealth really keeps creating happiness or if it reaches a plateau.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
poverty <- read_xlsx("data/povertybenelux.xlsx")
happiness <- read_xlsx("data/happinessbenelux.xlsx")
gdp <- read_xlsx("data/gdpbenelux.xlsx")
inflation <- read_xlsx("data/inflationbenelux.xlsx")
```

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

Steps taken for data cleaning

1. **strip non-numeric characters**
2. **pivot from wide to long**
3. **convert the year column to integer**
4. **have the GDP values also in line with the earlier observations that are already in billions**
5. **'left-join' all four long tables on 'country' and 'year'**

```
clean_long <- function(df, value_col) {
  df%>%
    pivot_longer(-Country,names_to = "year", values_to = value_col) %>%
    mutate(year = as.integer(year))
}
```

```
gdp <- gdp %>%
  mutate(across(-Country, ~ round(as.numeric(gsub(",", "", .)), 3)))

inflation <- inflation %>%
  mutate(across(-Country, ~ as.numeric(gsub("%", "", .))))

poverty <- poverty %>%
  mutate(across(-Country, ~ as.numeric(gsub("%", "", .))))
```

```r
gdp <- clean_long(gdp, "gdp")
inflation <- clean_long(inflation, "inflation")
happiness <- clean_long(happiness, "happiness")
poverty <- clean_long(poverty, "poverty")
```

```r
merged <- gdp %>%
  left_join(happiness, by = c("Country", "year")) %>%
  left_join(inflation, by = c("Country", "year")) %>%
  left_join(poverty, by = c("Country", "year"))
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

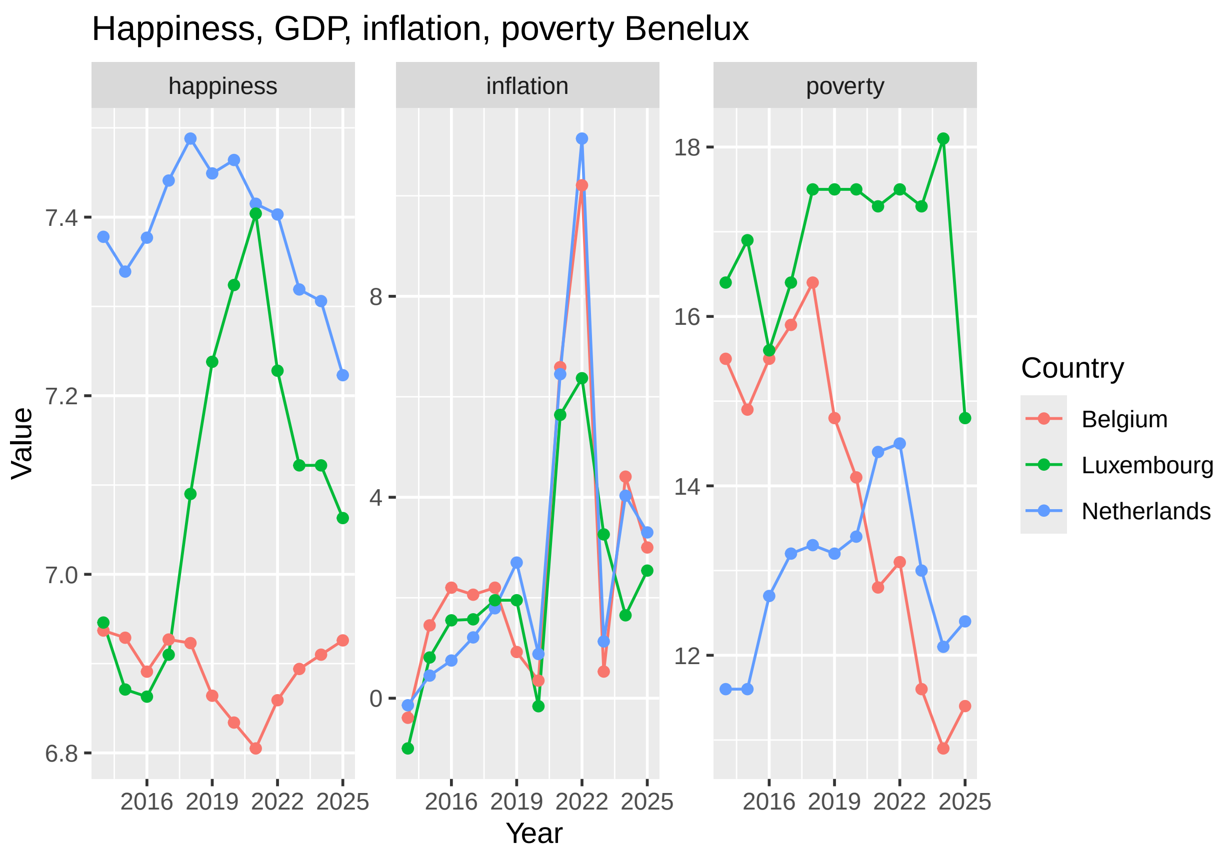
3.2 Generate necessary variables

Variable 1

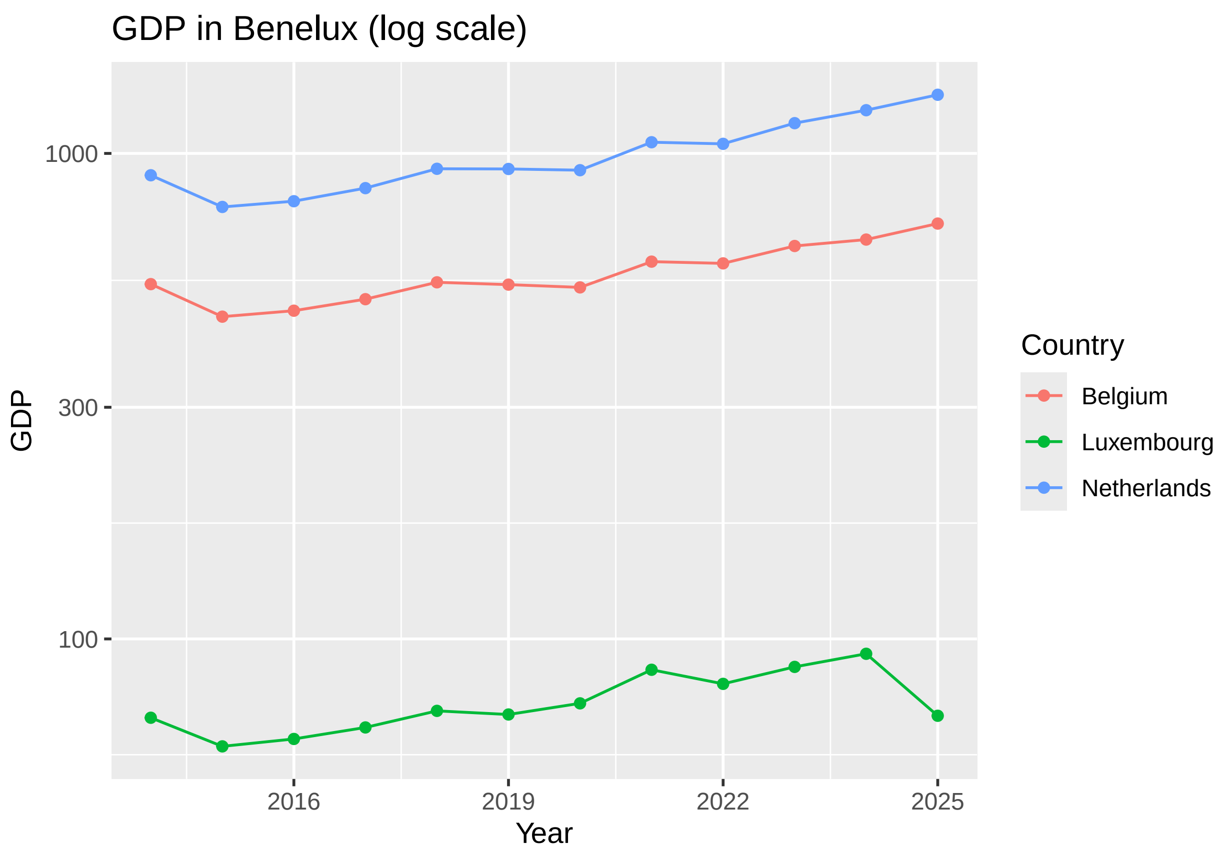
```r
merged <- merged %>%
  mutate(period = case_when(
    year <= 2017 ~ "Early",
    year <= 2021 ~ "Mid",
    TRUE ~ "Late"
  ))
```

3.3 Visualize temporal variation

```r
merged %>%
  pivot_longer(cols = c(happiness, inflation, poverty),
               names_to = "variable",
               values_to = "value") %>%
  ggplot(aes(x = year, y = value, color = Country)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ variable, scales = "free_y") +
  labs(title = "Happiness, GDP, inflation, poverty Benelux",
       x = "Year",
       y = "Value",
       color = "Country")
```




```r
ggplot(merged, aes(x = year, y = gdp, color = Country)) +
  geom_line() +
  geom_point() +
  scale_y_log10() +
  labs(title = "GDP in Benelux (log scale)",
       x = "Year",
       y = "GDP",
       color = "Country")
```



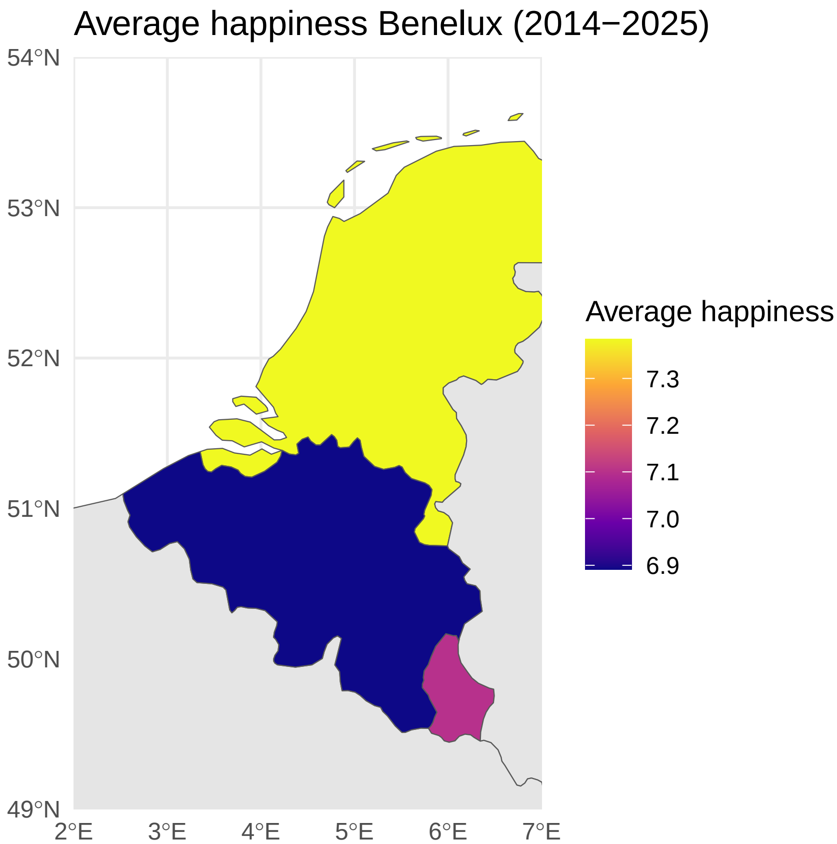
3.4 Visualize spatial variation

```
avg_happiness <- merged %>%
  group_by(Country) %>%
  summarise(avg_happiness = mean(happiness, na.rm = TRUE))
```

```
world_map <- ne_countries(scale = "medium", returnclass = "sf")
```

```
benelux_map <- world_map %>%
  left_join(avg_happiness, by = c("name" = "Country"))
```

```
ggplot(benelux_map) +
  geom_sf(aes(fill = avg_happiness)) +
  coord_sf(xlim = c(2, 7), ylim = c(49, 54), expand = FALSE) +
  scale_fill_viridis_c(option = "plasma", na.value = "grey90") +
  labs(title = "Average happiness Benelux (2014-2025)",
       fill = "Average happiness") +
  theme_minimal()
```

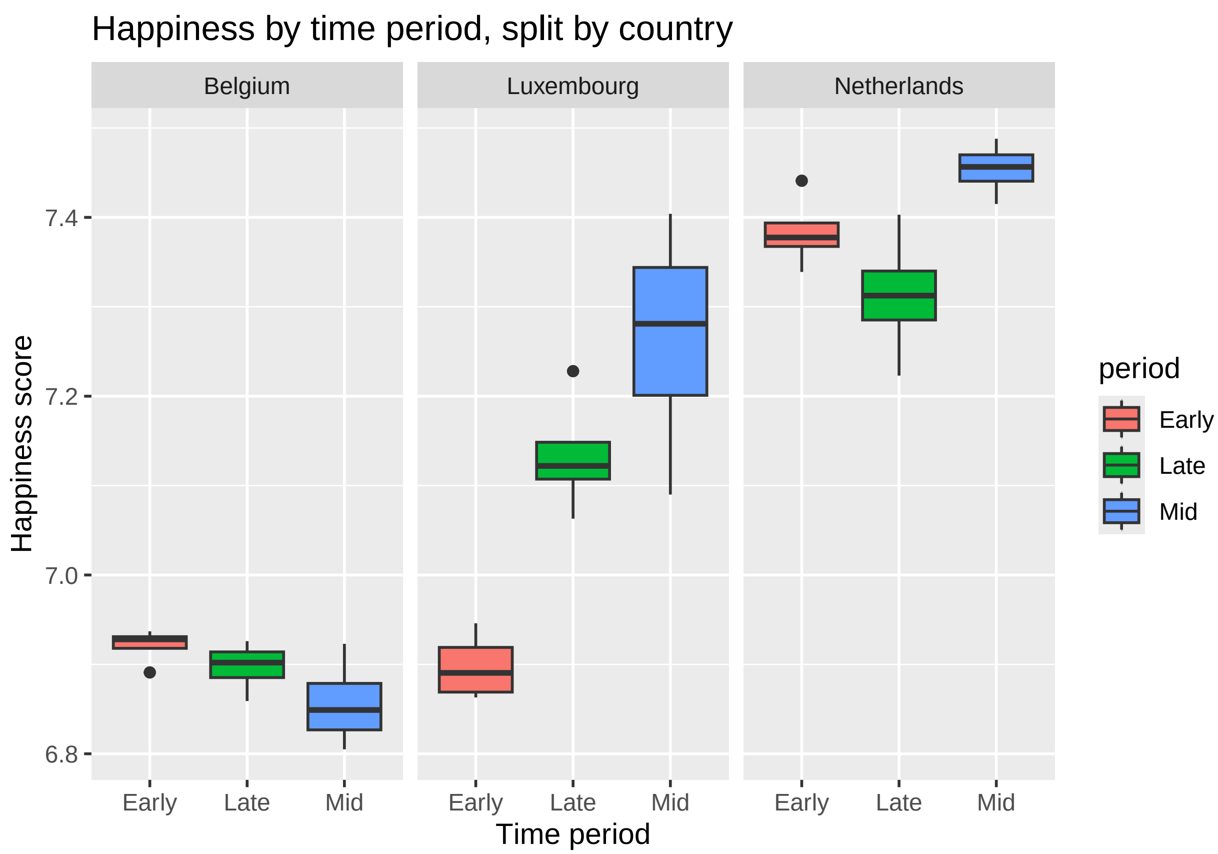


Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

```
ggplot(merged, aes(x = period, y = happiness, fill = period)) + geom_boxplot() +
  facet_wrap(~ Country) +
  labs(title = "Happiness by time period, split by country",
       x = "Time period",
       y = "Happiness score")
```

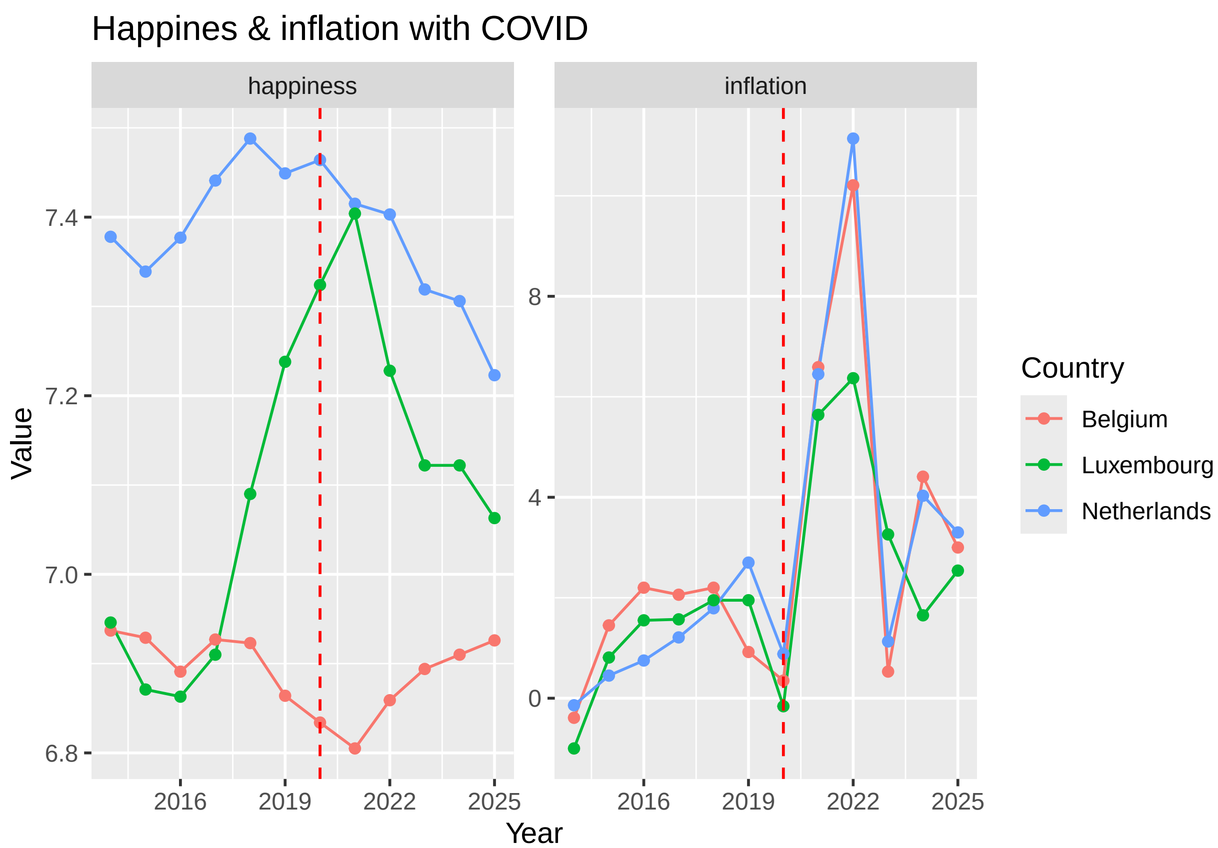


Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

```r
merged %>%
  pivot_longer(cols = c(happiness, inflation),
               names_to = "variable",
               values_to = "value") %>%
  ggplot(aes(x = year, y = value, color = Country)) +
  geom_line() +
  geom_point() +
  geom_vline(xintercept = 2020, linetype = "dashed", color = "red") +
  facet_wrap(~ variable, scales = "free_y") +
  labs(title = "Happines & inflation with COVID",
       x = "Year",
       y = "Value",
       color = "Country")
```



Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: . . .

5.2 Reference list

Use APA referencing throughout your document.

library(ggplot2)